Ashwin Padaki

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Summary

I use ideas and tools from theoretical computer science to develop algorithmic approaches to problems arising in machine learning and search systems. My work focuses on **nearest neighbor search**, **quantization**, and **attention**.

Education

University of Pennsylvania — Philadelphia, PA
Ph.D. in Computer and Information Science — 2024 – 2028 (expected)
Advisors: Erik Waingarten and Sanjeev Khanna.

Columbia University — New York, NY
B.A. in Computer Science, Mathematics — 2020 – 2024

Industry Experience

Research Scientist Intern — Jun 2026 – Aug 2026
Pinecone — New York, NY

- Built **VQ-bench**, an open-source, composable vector quantization benchmark.

Quantitative Trader Intern — Jun 2022 – Aug 2022
Optiver — Chicago, IL

Papers

Author order is alphabetical by default; otherwise, * denotes equal contribution.

1. **Attention via Black-Box Vector Search**
   Stepan Zharkov*, Krish Singal*, **Ashwin Padaki***, Alexandr Andoni
   Preprint, 2026
2. **A Near-Optimal Space Lower Bound for Euclidean Diameter Estimation in Dynamic Streams**
   Ashwin Padaki, Krish Singal, Erik Waingarten
   Preprint, 2026
3. **Prune, Don't Rebuild: Efficiently Tuning α-Reachable Graphs for Nearest Neighbor Search**
   Zachary Ives, Jiaming Liang, **Ashwin Padaki**, Erik Waingarten, Tian Zhang
   NeurIPS 2026
4. **VQ-bench: A Composable Vector Quantization Framework**
   Ashwin Padaki, Amir Ingber, Edo Liberty
   VecDB@VLDB 2026 · **Oral presentation**
5. **Learning Partition Trees for Nearest Neighbor Search**
   Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten
   VecDB@VLDB 2026
6. **Sparse Navigable Graphs for Nearest Neighbor Search: Algorithms and Hardness**
   Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten
   SODA 2026

Papers (continued)

7. **A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams**
   Sanjeev Khanna, **Ashwin Padaki**, Krish Singal, Erik Waingarten
   FOCS 2025
8. **Inapproximability of Maximum Diameter Clustering for Few Clusters**
   Henry Fleischmann, Kyrylo Karlov, Karthik C. S., **Ashwin Padaki**, Stepan Zharkov
   SODA 2025
9. **Smaller Low-Depth Circuits for Kronecker Powers**
   Josh Alman, Yunfeng Guan, **Ashwin Padaki**
   SODA 2023

Talks

VQ-bench: A Composable Vector Quantization Framework
VecDB@VLDB 2026 Workshop — Sep 2026

Learning Partition Trees for Nearest Neighbor Search
NYU Algorithms Seminar — May 2026

Sparse Navigable Graphs for Nearest Neighbor Search
SODA 2026 Conference — Jan 2026
NYC Graduate Student Theory Seminar — Nov 2025
DIMACS/Rutgers Theory Seminar — Nov 2025
CMU Theory Seminar — Sep 2025

Inapproximability of Maximum Diameter Clustering for Few Clusters
SODA 2025 Conference — Jan 2025

Reviewing

Conference reviewing: FOCS 2026; SODA 2027; SOSA 2027
Journal reviewer: ACM Transactions on Parallel Computing — 2025

Service

Head Teaching Assistant — Sep 2025 – May 2026
Algorithms for Big Data (UPenn)

Seminar Organizer — Sep 2025 – present
UPenn Theory Seminar

Seminar Instructor — Sep 2023 – Dec 2023
Columbia Undergraduate Learning Seminar in Theoretical Computer Science

Achievements

National Science Foundation (NSF) Graduate Research Fellow — 2024
Phi Beta Kappa Inductee — 2024
Putnam Mathematical Competition, Top 500 Scorer — 2022